

Passenger jets

Lining up on the runway at Gatwick Airport near London, UK, a passenger jet takes off every few minutes while others land nearby. Over a million people pass through the airport every year.

major investment. A world speed record was set when the French *train à grande vitesse* (TGV) reached speeds of 357 mph (575 km/h). These new trains promise to halve current journey times; a 315-mile (500-km) trip from Paris to Stuttgart will take just 3 hours and 40 minutes.

Advances in maritime engineering have produced faster, larger ships that are able to transport millions of tons of cargo around the world. Improved refrigeration techniques allow perishable foodstuffs to be carried by both sea and air, making products that were once a luxury readily available.

The development of cheaper and faster modes of travel has brought previously inaccessible parts of the world within reach of ordinary people. In the century following the Wright Brothers' first powered flight in 1904, aeronautical science has revolutionized aircraft design. Today, wide-bodied, high performance jet airliners carry millions of passengers every year.

Going places

Tourism is one of the world's most rapidly growing industries, driven by higher levels of disposable income in some societies, as well as developments in technology and the transport infrastructure.

Taking vacations became the social norm in the developed world as terms of employment began to include periods of paid leave in the early decades of the 20th century. At first, people tended to vacation in domestic locations, but from the 1950s, mass tourism, particularly between European destinations, became common. Long-haul travel between

"... people's lives change so fast, that a person is **born into one kind of world** [and] **grows up in another...**"

MARGARET MEAD, US ANTHROPOLOGIST

continents remained expensive until the 1980s, when improved airline passenger capacity and shortened journey times

led to an increase in long-haul traffic. This upward trend has continued, with growing numbers of people traveling between continents for their vacations. In recent years, tourism has become

INDUSTRIALIST (1863–1947)

HENRY FORD

Henry Ford founded the Ford Motor Company in 1903. From 1909, he began mass production of the Model T Ford, using assembly line methods and wage incentives. Soon the company was producing a complete car in 1 hour and 33 minutes and its price fell from \$1,000 to \$360. Ford became the world's leading car manufacturer. He opposed World War I and sponsored an international conference to try to negotiate an armistice. In the 1930s, he was violently opposed to unions and was the last car manufacturer to recognize union workers in his factories.



Satellite communications

A Western journalist transmits a report by satellite phone from the hills of Northern Afghanistan. Portable satellite communications devices can provide news of events from around the world as they occur.

more specialized. With the growth in "niche markets" catering for special interests such as extreme sports, the market has both grown and fragmented. Destinations that were once the preserve of intrepid explorers, such as the Galápagos Islands (see pp.340–41), are now tourist hotspots. Nowhere is inaccessible—even Mount Everest, the summit of which was only reached for the first time by Edmund Hilary and Tenzing Norgay in 1953, has become a tourist destination.

Instant communication

The technological revolution gathered pace from the late 1950s with the invention of the microchip (see pp.460–61). This transformed the modern world. Microtechnology has touched most fields of manufacturing and engineering, from weapons design to medical robotics. The impact on human communications has been huge. Microchips provide the "brains" for computers, personal communicators, and cellular phones, allowing instant global communication and the storage of vast amounts of information. Data that previously filled libraries can now be stored on one tiny chip. The creation of the Internet (see right) allowed this information to be shared among computer users worldwide.

It is often said that the 21st century will be the "Communications Age",

The seemingly insatiable public appetite for travel to exotic locations shows no sign of fading in the 21st century. Not content with gaining access to the remotest corners of Earth, there is now a new demand for vacations that are literally out of this world.

SPACE TOURISM

The British entrepreneur Richard Branson has poured millions of dollars into the Virgin Spaceship to take passengers into space. Wealthy individuals and corporations are already booking "personal spaceflights" with the Russian Space Agency. Tourists have to undergo rigorous physical training and medical tests before being declared fit to travel. To date, five people have vacationed in space, paying as much as \$25 million for the privilege. The first tourist, California financier Dennis Tito, described his eight-day voyage as "a trip to paradise."

an age centered on shared information between individuals, businesses, and nations. In the past, geographical distance impeded communication, but technology has broken down these barriers. The Communications Age

2 BILLION people worldwide regularly access the Internet.

offers the chance for knowledge to be shared across the world. After more than a century in which international conflicts have wrought so much carnage, some have begun to promote a new ideal for this age: that of the global citizen.

INVENTION

THE WORLD WIDE WEB

British physicist Tim Berners Lee invented the World Wide Web in the late 1980s and early 1990s. Lee developed the concept of "hypertext" in order to share research with fellow scientists across a single computer network. His original ideas led to the popularization of the Internet as people used this technology to build websites, which could be accessed from computers anywhere in the world. Lee currently heads the World Wide Web consortium that concentrates on improving Internet technology and accessibility.



Personal communicators

These video phones are the latest development in cell phone technology. Utilizing digital multimedia satellite broadcasting, the video phones broadcast news, music, and drama channels.



Personal computers

One of the first personal computers was this Sinclair ZX Spectrum, which was used in association with an ordinary TV set. This made it cheap and accessible for everyone to have their own computer.



Global Positioning System

This GPS receiver picks up signals transmitted by global navigation satellites to determine its location and direction. GPS technology has become an important tool for mapmaking, land surveying, and navigation.